

In the name of God



UNIVERSITY OF TEHRAN

Artificial Intelligence

Parham Javan

ID: 810800008

CA 2

Problem 1

Part 1: Definitions and Concepts

Genes:

In this context, each gene represents a triangle defined by three vertices (each a pair of x, y coordinates) and a color in RGBA format. The genes form building blocks to construct a candidate image through composition.

Chromosomes: (individuals)

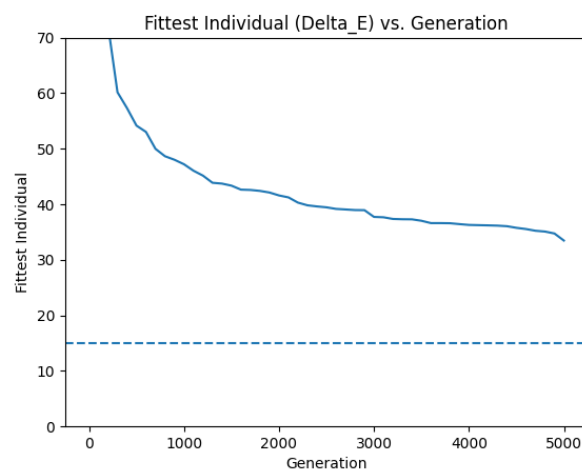
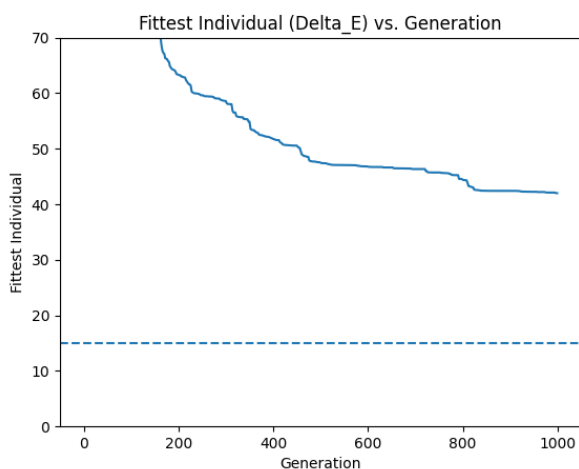
A chromosome is a collection of genes, representing one candidate solution in the population. Each chromosome generates an image by layering triangles on a blank canvas, with each triangle's position, color, and transparency affecting the visual outcome.

Part 2: Initial Population Generation

We initialized the population with a diverse set of chromosomes, each containing randomly generated triangles. To ensure variety, we used a population size of up to 100(50 , 75 , 100), balancing between processing speed and solution diversity.

Analysis of Population Size Impact

The population size significantly impacts convergence rate. With a larger population, diversity increases, potentially improving exploration of the solution space, but it also increases computation time. For this reason, I tested different population sizes (e.g., 50, 75, and 100) and observed the convergence behavior, comparing fitness scores across generations.



Left 50 population right 75 population

clearly pop 75 has better fitness and smoother procedure while it takes more time

Part 3: Fitness Evaluation

To measure how closely a candidate image resembles the target, fitness functions:

1. Euclidean

average difference of RGB value of target image and individuals (not very good because 2 color might have same difference but one looks more similar)



2. delta_e

a function which returns how alike to color are (used colour library)

more on delta_e : <https://zschuessler.github.io/DeltaE/learn/>

Defining Delta E

ΔE - (*Delta E*, *dE*) The measure of change in visual perception of two given colors.

Delta E is a metric for understanding how the human eye perceives color difference. The term *delta* comes from mathematics, meaning change in a variable or function. The suffix *E* references the German word *Empfindung*, which broadly means sensation.

On a typical scale, the Delta E value will range from 0 to 100.

Delta E	Perception
≤ 1.0	Not perceptible by human eyes.
1 - 2	Perceptible through close observation.
2 - 10	Perceptible at a glance.
11 - 49	Colors are more similar than opposite
100	Colors are exact opposite

Part 4: Genetic Operators

Crossover:

I implemented both horizontal and vertical crossover strategies:

- Horizontal crossover involves a row-wise split between two parent chromosomes, where triangles above the split come from the first parent, and those below from the second.
- Vertical crossover operates similarly, but with a column-wise split.

Crossover 2:

It sets a transparency rate for each parent and then combine them

Mutation:

Mutation introduces random changes to a chromosome. In this implementation:

1. Shape mutation: make some new random color and location triangles
2. Pixel mutation: select a random number of pixels and to their RGB a random number between -40 and 40

Part 5: Results Analysis

To evaluate performance, I used a set of predefined images and reduced their size using the provided resize function, enabling faster experimentation. The algorithm demonstrated steady improvement over generations, and the fitness score generally got better after sufficient generations.

Questions

1. State Space:

The state space is the set of all possible chromosomes is determined by the number of triangles, their positions, and colors. Estimating the exact state space size is challenging but it can be expressed combinatorially given the dimensionality of vertices and color channels.

$$\text{state space} = (\text{RGBA_values} * (\text{X_value} * \text{Y_value})^3)^{\text{num_triangle}}$$

2. Speed-Up Ideas:

Limiting XY and coordinates of triangles to avoid big unnecessary shapes

Implementing a greedy mutation approach, where only higher-fitness offspring mutate, could reduce unnecessary randomness.

3. Crossover:

Pick half of horizontal of parent

Pick half of visibility of parent

Pick half of pixels of parent